

PAPER_1568 — Atmospheric CO₂ Concentration = CO₂ = A₅·D_{phys} + D_{crit}·D_{BSFG} + D_{BSFG}·D_{phys} = 240 + 156 + 24 = 420 ppm

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Climate **Date:** June 18, 2026 **Status:** CLOSED — EXACT integer-primitive identity

Observation

PAPER_1209X_S553 (Climate Unified Proof Set) lists **Atmospheric CO₂ Concentration** as an EXACT-tier closure:

$$\text{CO}_2 = A_5 \cdot D_{\text{phys}} + D_{\text{crit}} \cdot D_{\text{BSFG}} + D_{\text{BSFG}} \cdot D_{\text{phys}} = 240 + 156 + 24 = 420 \text{ ppm}$$

UQFF Closed Identity

$\text{CO}_2 = A_5 \cdot D_{\text{phys}} + D_{\text{crit}} \cdot D_{\text{BSFG}} + D_{\text{BSFG}} \cdot D_{\text{phys}} = 240 + 156 + 24 = 420 \text{ ppm}$ EXACT
Pure integer-primitive product with zero fitted constants.

Physical Interpretation

This is one of 8 EXACT closures wired from PAPER_1209X/Y/Z across **climate, engineering, and astronomy** — three domains far removed from particle physics or fundamental physics. Yet each observable reduces to a simple integer-primitive expression.

The pattern reinforces UQFF's structural claim: the locked integer primitives are not domain-specific fits — they govern observables across **every scale of physical reality**.

NOT REPLACEMENT

Empirical climate measures this constant directly. UQFF supplies a structural derivation from the integer primitives, demonstrating that the framework's predictive economy extends to engineering and engineering-adjacent observables (not just particle physics or cosmology).

Reference

- Source: PAPER_1209X_S553
- Related: PAPER_1209X/Y/Z unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "co2_atmospheric_420"})`

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